

Questions on Conditional Statements

1. Write a program to get the number of days in the month entered by the user.
2. Write a program that takes marks (out of 100) as input from the user and assigns grades:
 - 90 and above: Grade A
 - 80–89: Grade B
 - 70–79: Grade C
 - 60–69: Grade D
 - Below 60: Fail
3. Write a program that takes the age of a person as input and checks if they are eligible to vote (18 years or older).
4. Write a program that takes the x and y coordinates of a point and determines the quadrant it belongs to.
5. Write a program that takes the number of electricity units consumed and calculates the bill:
 - ≤ 100 : Rs. 5/unit
 - 101–200: Rs. 8/unit
 - 200: Rs. 10/unit
6. Write a program that takes a person's citizenship and passport validity (in years) and checks if they are eligible to travel internationally.
7. Write a program that takes annual income and calculates tax:
 - $\leq 2,50,000$: No tax
 - 2,50,001–5,00,000: 5%
 - 5,00,001–10,00,000: 20%
 - 10,00,000: 30%
8. Write a program that takes a student's marks and household income and determines if they are eligible for a scholarship.
9. Write a program that takes weight (kg) and height (m), calculates BMI, and classifies as:
 - < 18.5 : Underweight
 - 18.5–24.9: Normal
 - 25–29.9: Overweight
 - ≥ 30 : Obese

10. Write a program that takes age and determines ticket price:

- < 5: Free
- 5–17: Rs. 100
- 18–60: Rs. 200
- 60: Rs. 150

11. Write a program that takes temperature (°C) and categorizes it as:

- Below 0: Freezing
- 0–10: Very Cold
- 11–20: Cold
- 21–30: Moderate
- 30: Hot

12. Write a program that takes a month number (1–12) and determines the season:

- Mar–May: Spring
- Jun–Aug: Summer
- Sep–Nov: Autumn
- Dec–Feb: Winter

13. Write a program that takes hardness, carbon content, and tensile strength and determines the grade.

14. Write a program that takes item price and applies discount:

- < 1000: No discount
- 1000–5000: 10%
- 5000: 20%

15. Write a program that takes a year and checks if it is a century year (divisible by 100).

16. Write a program that takes distance (km) and calculates taxi fare:

- First 5 km: Rs. 50/km
- Next 15 km: Rs. 40/km
- Beyond 20 km: Rs. 30/km

17. Write a program that takes a number and checks if it is even or odd.

18. Write a program that takes a year and checks if it is a leap year.

19. Write a program that takes three numbers and prints the largest.
20. Write a program that checks login by asking for username and password.
21. Write a program that takes three sides and checks if the triangle is valid.
22. Write a program that checks whether a given character is a vowel or consonant.
23. Write a program that checks if a number is divisible by 3, 5, or both.
24. Write a calculator program that performs $+$, $-$, $\times$, $\div$ based on user input.
25. Write a program that checks if a number is prime.
26. Write a program that checks whether a character is an alphabet, digit, or special character.
27. Write a program to classify triangle as Equilateral, Isosceles, or Scalene.
28. Write a program that takes GPA (out of 10) and gives a letter grade.
29. Write a program that takes a traffic signal color and prints Stop/Wait/Go.
30. Write a program that calculates hotel cost and applies discount:
 - < 3 nights: No discount
 - 3–5 nights: 10%
 - 5 nights: 20%
31. Write a program that calculates water bill based on usage:
 - $\leq 100\text{L}$: ₹2/L
 - 101–300L: ₹4/L
 - 300L: ₹5/L
32. Write a program to check if student is eligible for exam (attendance $\geq 75\%$).
33. Write a program to calculate parking fee:
 - First 2 hrs: ₹20/hr
 - Next 3 hrs: ₹15/hr
 - Beyond 5 hrs: ₹10/hr
34. Write a program that takes a month number (1–12) and prints number of days.
35. Write a program that classifies person as child, teenager, adult, or senior.
36. Write a program that checks whether a number is positive, negative, or zero.